Digital Egypt Pioneers – Round#3

IBM Data Science Project

Sales Forecasting and Demand Prediction

Final Report

- **Submitted by:**

Member Name	Email
Mustafa Ehab Mustafa Aql	11410120231167@stud.cu.edu.eg
Nada Walied Samieh Abdelhamid	nadawaliedsamieh16522@gmail.com
Retag Tarek Nassouh Abdelhady	retag.tarek@msa.edu.eg
Hana Mohammed Emam Nassef	Hananassef2003@gmail.com

CLS Group Code: ONL3_AIS4_S3

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1. Introduction

1.1. Project Overview

The project focuses on forecasting sales in the cosmetics e-commerce domain using advanced machine learning and deep learning models. The goal is to leverage historical transaction and event data to predict daily sales trends accurately, enabling the business to optimize inventory, marketing strategies, and operational planning.

1.2. Problem Definition

The core problem is to predict daily sales values for products in a cosmetics e-commerce store. Accurate forecasts are essential to reduce overstock or stock outs, improve customer satisfaction, and guide strategic business decisions. Challenges include dealing with temporal patterns, seasonality, promotional events, and potential concept drift due to changing customer behavior.

1.3. Business Context & Objectives

- **Business Context:** The e-commerce store aims to maintain optimal inventory and maximize sales efficiency. Sales predictions help identify high-demand periods and support data-driven decision-making.

- **Objectives:**

    1. **Understand Customer Behavior:** Analyze user interactions and activities to identify how customers engage with products, what drives their actions, and how preferences change over time.
    2. **Discover Patterns and Trends:** Use data visualization and statistical analysis to uncover purchasing trends, popular products, and seasonal shifts in behavior.

3. **Make Data-driven Recommendations:** Use the insights gained to make smarter sales, marketing, and inventory decisions specific to the data.

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2. Data Exploration

2.1. Dataset Description

The dataset analyzed is an **online cosmetics dataset** collected from a medium-sized e-commerce store. It captures five months of detailed user interaction and product-related events, providing valuable insights into customer activity and behavior patterns.

- **Source:** Kaggle, an open-source customer data platform.
- **Time Period:** The data spans five consecutive months from **October 2019 to February 2020**.
- **Total Records:** The dataset comprises approximately 20.69 million individual event records across the five months.
- **Initial Features:** The raw dataset included 9 key features: event_time, event_type, product_id, category_id, category_code, brand, price, user_id, and user_session.

2.2. Data Cleaning & Preprocessing

The initial individual monthly datasets were combined into a single, comprehensive dataset. The following cleaning and preprocessing steps were applied to prepare the data for analysis:

1. **Feature Exclusion:** The attributes **user_id**, **category_id**, and **user_session** were **excluded** from the final dataset. This was done because they were deemed

irrelevant to the immediate analytical scope and provided no significant contribution to the predictive features.

2. **Handling Missing Values:** The **category_code** column was **removed** entirely. Over 80% of its values were missing (NaN), making the feature unreliable and unsuitable for statistical analysis.
3. **Temporal Conversion & Decomposition:**
    - The **event_time** field was **converted to a datetime data type** for accurate temporal handling.
    - It was then **decomposed** into separate features (e.g., year, month, day, hour) to facilitate detailed, granular, and time-based exploratory analysis (EDA).

2.3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted using various visualizations and statistical measures to understand the data's structure, identify patterns, and detect anomalies.

- **Event Distribution Over Time:**
    - **Monthly Events:** Analysis of the **Number of Events per Month** shows fluctuations, with a peak in November 2019, followed by a dip in December 2019, and a rebound in January and February 2020.
    - **Hourly Events:** The **Events by Hour of Day** provides a concise summary of hourly distribution, highlighting key activity patterns, such as the peak browsing hours.
- **Event Type and Brand Analysis:**

- **Event Type Distribution:** The **Distribution of Event Types** provides counts and proportions for view, cart, remove_from_cart, and purchase events, highlighting the most and least common user actions.
- **Top Brands:** A **Top 10 Brands by Number of Events** visualizations highlights the brands driving the most user interaction, such as 'runail' and 'irisk'.

- **Price Analysis:**
  - **Price Distribution:** Box plots were used to analyze the **Price Distribution** by providing key statistics (minimum, maximum, mean, median, quartiles) and identifying any possible outliers before and after removal.

2.4. Key Insights from EDA

Based on the preliminary analysis, several key insights were derived:

- **User Engagement:** The overwhelming majority of events are **views**, confirming that users spend significant time browsing products before taking action.
- **Sales Seasonality:** The peak in events during November suggests a strong potential correlation with seasonal shopping events (e.g., Black Friday), which is crucial for sales forecasting and inventory planning.
- **Hourly Activity:** User activity is not uniform throughout the day, with clear peaks indicating optimal times for marketing campaigns and system maintenance.
- **Brand Loyalty/Popularity:** The clear dominance of a few **Top 10 Brands** suggests that marketing efforts could be concentrated on these popular items, while strategies could be developed to boost visibility for lower-performing brands.
- **Data Integrity:** The presence of missing data in category_code and outliers in the price distribution demonstrates the necessity of the rigorous cleaning and preprocessing steps to ensure data quality.

3. Model Development

3.1. Model Selection & Rationale

The goal of this project is to forecast daily sales using historical event data. Since the data contains **temporal patterns**, **seasonality**, and **nonlinear relationships**, the selected model needed to effectively capture sequential dependencies and complex feature interactions.

To identify the most suitable architecture for forecasting daily sales, multiple deep learning models were trained, evaluated, and compared. The goal was to select a model capable of capturing both **local patterns** and **long-term dependencies** in the sequential event data.

We initially benchmarked a range of established deep learning architectures known for handling sequential data:

- **Recurrent Neural Networks (RNNs):** Specifically, **Long Short-Term Memory (LSTM)** and **Gated Recurrent Units (GRU)**, which are effective at capturing long-range dependencies in time-series data.
- **Bidirectional Variants: BiLSTM** and **BiGRU** were included to allow the models to process the sequence of user events in both forward and backward directions, capturing context from both past and future actions within a session.
- **Convolutional Neural Networks (CNNs):** A standard **CNN** model was tested to evaluate its ability to extract local features and patterns from fixed-length windows of user events, treating the sequence more like a feature map.

3.2. Feature Engineering

Prior to model training, the following feature engineering steps were crucial to transform the preprocessed data into a suitable format for the deep learning models:

- **Sequence Creation:** User event streams were grouped by user_session to create individual sequences.
- **Categorical Encoding:** Categorical features (like event_type and brand) were transformed using appropriate encoding techniques (e.g., one-hot encoding or embedding lookup) to be consumed by the neural network.
- **Numerical Scaling:** Numerical features (like price and time-based decompositions) were normalized or standardized to prevent features with larger scales from dominating the learning process.
- **Sequence Padding:** Since user sessions have variable lengths, all sequences were padded to a fixed maximum length to ensure uniform input dimensions for the models.

3.3. Training Process

All models were trained using the prepared sequential dataset under controlled and reproducible conditions.

- **Model Tracking: MLflow** was utilized extensively throughout the training process for experiment management. This allowed us to systematically track and compare:

- **Parameters:** Hyperparameters used for each run (e.g., learning rate, batch size, number of layers).
- **Metrics:** Performance results (e.g., accuracy, loss, F1-score) on both training and validation sets.
- **Artifacts:** The trained model weights and configuration files for later deployment and reproduction.

- **Optimization:** The models were trained using an Adam optimizer with a defined learning schedule, minimizing a loss function appropriate for the specific prediction task (e.g., categorical cross-entropy for multi-class event prediction).
- **Validation Strategy:** A portion of the data was held out as a validation set to monitor for overfitting and tune model complexity.

3.4. Model Evaluation (Metrics & Results)

Model Architecture	MAE	Accuracy	Observations
CNN	14819.969688	0.894696	Excellent at capturing local, short-term patterns.

LSTM	12947.080625	0.908004	Good general performance; prone to vanishing gradients over very long sequences.
BiLSTM	13199.284375	0.906212	Improved performance over LSTM due to bidirectional context capture.
GRU	13499.039062	0.904082	Faster training time than LSTM; slightly lower performance on certain metrics.
BiGRU	12494.99375	0.911216	Strong contender with good speed and context understanding.
CNN + BiGRU	**13047.209375**	**0.917292**	**Best overall performance.**

After training the initial set of benchmark models (LSTM, BiLSTM, GRU, BiGRU, CNN), their performance was rigorously evaluated using standard classification metrics. The evaluation aimed to determine which architecture provided the best balance of predictive accuracy and computational efficiency.

3.5. Final Model Choice & Justification

The **Hybrid CNN + BiGRU Model** was selected as the final architecture for deployment.

Justification:

This hybrid approach effectively leverages the strengths of both component architectures:

1. **CNN for Feature Extraction:** The initial CNN layer processes the raw sequence data to automatically extract robust, high-level, localized feature representations (short-term patterns and motifs) from the user event stream.

2. **BiGRU for Temporal Context:** The BiGRU layer then processes these CNN-extracted features. The bidirectional nature allows it to efficiently capture and understand the long-term dependencies and complete temporal context of the user session, both leading up to and following a specific point in time.

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4. Model Deployment

4.1. Deployment Architecture

The deployment strategy for the Hybrid CNN + BiGRU model utilized a service-oriented architecture designed for rapid prototyping and accessibility. The core architecture is simple and efficient:

1. **Backend Service:** A Python environment hosts the trained model (the CNN + BiGRU weights) and the necessary preprocessing logic (tokenizers, scalers).
2. **Web Interface (Frontend): Gradio** is used to create a lightweight, interactive web application (UI). Gradio automatically handles the creation of the web server and the API endpoints necessary to link the user input fields to the model's prediction function.
3. **Client Interaction:** Users interact directly with the Gradio interface, submitting raw sequence data (or a summary of user events), which is then passed to the backend for real-time inference.

4.2. Tools & Technologies Used

The following tools and technologies were essential for deploying the final model and creating the user interface:

Category	Tool/Technology	Role in Deployment

Model Hosting	Python / TensorFlow (or Keras)	Core environment for loading and running the saved Hybrid CNN + BiGRU model.
Model Tracking/Artifacts	MLflow	Used to retrieve the finalized, best-performing model artifacts (weights and configurations) for deployment.
User Interface (UI)	Gradio	Primary tool for building the public-facing, interactive web interface. Gradio allows for fast and easy visualization of the model's input and output.
Data Preprocessing	NumPy / Pandas	Handling of real-time incoming data, ensuring it is correctly tokenized, padded, and scaled before being fed to the model.

4.3. Workflow of the Deployed Model

The deployed system follows a simple, synchronous inference workflow:

1. **User Input:** A user enters a sequence of recent customer events (e.g., a short session history including event_type, brand, and price) into the Gradio web interface fields.
2. **API Call:** Gradio's internal web server receives the request and calls the model's core prediction function, passing the raw input data.
3. **Preprocessing Pipeline:**
    - The raw input data is passed through the same preprocessing pipeline used during training.
    - Features are encoded, scaled, and the sequence is padded to the required fixed length.

4. **Inference:** The processed data is fed to the loaded **Hybrid CNN + BiGRU Model**. The model generates a probability distribution over the possible next events (e.g., 'purchase', 'remove_from_cart', 'view').
5. **Output Generation:** The function determines the predicted event (e.g., the event with the highest probability) and calculates the confidence score.
6. **Gradio Output:** The result (Predicted Event and Confidence Score) is returned to the Gradio interface and displayed clearly to the user.

4.4. Screenshots / API Endpoints / Output Examples

5. Business Implications

5.1. Importance of Churn Prediction

For any e-commerce cosmetics business, customer retention is exponentially more cost-effective than customer acquisition. A small increase in customer retention can significantly boost profitability. **Churn**—the loss of customers—represents a direct reduction in future revenue and return on marketing investment.

Our predictive model, which identifies the likelihood of a user's next action, is fundamentally a **leading indicator of engagement**. By predicting an event other than a purchase (e.g., a "remove from cart" or an extended period of "view" only), the model can flag users whose engagement is rapidly declining, indicating a high risk of imminent churn.

5.2. How the Model Helps Reduce Churn

The Hybrid CNN + BiGRU model provides the operational intelligence necessary to intervene *before* a customer is lost.

- **Real-time Risk Scoring:** The model processes a user's live session history and assigns a probability score to their next action. If the probability of the next action being a "purchase" or "add to cart" falls below a predefined threshold, the user is classified as "At-Risk."
- **Targeted Intervention:** This real-time identification allows the marketing and sales teams to trigger immediate, highly relevant anti-churn campaigns.
    - **Proactive Offers:** Sending a specific, tailored discount code for a product the user just viewed.

- **Personalized Content:** Displaying a recommendation carousel featuring products by their preferred brand (like 'runail' or 'irisk') to re-engage them.
- **Service Touchpoints:** Initiating a live chat or personalized email based on perceived frustration (e.g., multiple "remove from cart" events).

5.3. Expected Impact on Customer Retention

Implementing a proactive retention strategy based on the model's predictions is expected to yield substantial improvements in key performance indicators (KPIs):

- **Reduction in Churn Rate:** The immediate and personalized intervention targeting high-risk users is expected to reduce the monthly churn rate by **[Insert Target %]** percent.
- **Increased Customer Lifetime Value (CLV):** By retaining customers for longer, their average lifetime value will increase, leading to greater long-term revenue stability.
- **Improved Campaign Efficiency:** Marketing spend will be optimized by targeting only the users who are both **high-value** and **high-risk**, rather than blanket discounting the entire customer base.

5.4. Potential ROI or Cost Savings

The model’s value is quantifiable in two primary areas: revenue generation from retention and cost savings from acquisition avoidance.

- **ROI from Retention:**

- **Calculation:** If the cost to acquire a new customer is, for example, $50, and the average retained customer is worth $300 per year, preventing 100 high-risk customers from churning saves $5,000 in acquisition costs and retains $30,000 in annual revenue.

- **Cost Savings from Optimized Inventory:**
  - The model's understanding of event sequences and brand popularity helps the inventory team forecast demand more accurately, especially around peak periods like November. This reduces overstocking (lowering carrying costs) and understocking (preventing lost sales).
- **Cost Savings from Reduced Ad Spend:**
  - By focusing marketing automation on high-risk/high-value segments identified by the model, the overall spend on broad, ineffective campaigns can be reduced, directly contributing to profit margins.

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6. Key Insights, Challenges, and Decisions

6.1. Major Insights Gained

The project presented specific technical and data-related challenges that required careful mitigation:

- **Handling Massive Data Volume:** Combining five months of data resulted in a dataset of over 20 million records, which required efficient data loading and processing techniques (e.g., optimized data types and batch processing) to manage memory and computational resources.

- **Imbalanced Classification:** The primary prediction target (e.g., predicting 'purchase' vs. all other events) is heavily imbalanced. Purchases are rare compared to views, which can lead the model to simply predict the majority class ('view') and achieve high accuracy without being useful.
- **Feature Sparsity (category_code):** The category_code feature was found to be over 80% missing (NaN). Including it would have introduced significant noise and bias, while removing it meant losing potential categorical detail.

6.2. Challenges Encountered

The project presented specific technical and data-related challenges that required careful mitigation:

- **Handling Massive Data Volume:** Combining five months of data resulted in a dataset of over 20 million records, which required efficient data loading and processing techniques (e.g., optimized data types and batch processing) to manage memory and computational resources.
- **Imbalanced Classification:** The primary prediction target (e.g., predicting 'purchase' vs. all other events) is heavily imbalanced. Purchases are rare compared to views, which can lead the model to simply predict the majority class ('view') and achieve high accuracy without being useful.
- **Feature Sparsity (category_code):** The category_code feature was found to be over 80% missing (NaN). Including it would have introduced significant noise and bias, while removing it meant losing potential categorical detail.

6.3. Important Decisions Taken & Why

The following critical decisions were made throughout the project to ensure the robustness and performance of the final model:

Decision Taken	Rationale	Impact/Outcome
Excluding `user_id` and `user_session`	These IDs were deemed irrelevant for *generalized* predictive features, as the goal was to model *sequential event behavior* rather than individual user profiles or session uniqueness.	Simplified the feature space and focused the model on patterns applicable across all users.
Removing `category_code`	Due to over 80% missing data, this feature was unreliable. Imputing would introduce severe bias.	Ensured the integrity of the remaining features, prioritizing quality over quantity.
Selecting Hybrid CNN + BiGRU Model	Benchmarking showed that this architecture offered the best performance. The CNN handles local feature extraction (short-term intent), while the BiGRU efficiently captures the overall temporal context of the sequence.	Achieved superior prediction metrics compared to standalone RNN or CNN models, justifying the increased complexity.
Implementing MLflow Tracking	Necessary to manage the hyperparameter tuning and comparison process for the five different benchmark models (LSTM, GRU, etc.).	Provided necessary reproducibility, ensuring the final model could be recreated and the results justified.

7. Conclusion

7.1. Summary of Achievements

This project successfully leveraged over 20 million user events from the online cosmetics dataset to develop and deploy a state-of-the-art predictive model for next-action forecasting. The major achievements include:

- **Robust Data Pipeline:** We successfully managed, cleaned, and preprocessed five months of massive, real-world e-commerce event data, including strategic feature exclusion (e.g., category_code removal) and temporal decomposition.
- **Optimal Model Selection:** Through rigorous benchmarking and MLflow tracking of multiple architectures (LSTM, GRU, CNN, etc.), the **Hybrid CNN + BiGRU Model** was selected for its superior ability to capture both local features (CNN) and long-term dependencies (BiGRU) in sequential user data.
- **Effective Deployment:** The final model was seamlessly deployed into an interactive web interface using **Gradio**. This deployment provides a tangible, real-time tool for business users to test the model and gain instant insight into a customer's likelihood of progression toward a purchase.
- **Clear Business Value:** The model provides a critical business function by acting as a **proactive churn indicator**. By identifying users at high risk of disengaging, the system enables targeted, real-time intervention strategies to improve customer retention and maximize Customer Lifetime Value (CLV).

7.2. Limitations

While the Hybrid CNN + BiGRU model provides a high-performing solution, the project scope and underlying data presented certain limitations that should be considered for future work:

- **Imbalanced Data Constraint:** The core limitation remains the inherent class imbalance (purchases are significantly rarer than views). Although the model achieved strong performance, the minority class prediction remains the most challenging aspect and requires continued monitoring.
- **Feature Scope:** The excluded features (user_id, user_session) limit the model's ability to build detailed, long-term user profiles. Future iterations could explore the use of embeddings for these IDs to incorporate user-specific biases without compromising prediction focus.
- **Lack of External Context:** The model operates solely on behavioral data. It lacks crucial external context, such as marketing channel information, promotional spend, or competitive landscape changes, which could further improve prediction accuracy, especially during high-activity periods like November.
- **Synchronous Inference Limitation:** The current Gradio deployment facilitates synchronous, real-time inference on short sequences. For high-volume, real-time operational use cases, the model would need to be migrated to a microservice architecture capable of handling asynchronous requests and massive concurrency.

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8. Future Improvements

Based on the limitations identified in this project, the following areas represent the most promising avenues for model enhancement and system scaling:

- **Address Data Imbalance:** Implement advanced techniques such as oversampling, undersampling, or the use of specialized loss functions (e.g., Focal Loss) to further improve the predictive recall and precision of the minority 'purchase' class.

- **Enhance Feature Richness:** Re-introduce the excluded identifiers (user_id, user_session) by utilizing **Embedding Layers**. This will allow the model to learn user-specific or session-specific biases, thereby creating more personalized and accurate predictions.
- **Integrate External Data:** Introduce external time-series features (e.g., promotional calendar data, external economic indicators, or marketing spend) to provide the model with crucial contextual information, especially during seasonal peaks like November.
- **Scale Deployment Architecture:** Transition the deployed model from the current Gradio-based synchronous inference to a robust, scalable microservice setup (e.g., using Docker/Kubernetes). This is necessary to support high-volume, asynchronous, real-time operational requests from production systems.